

AUTOMATED REVIEW RATING SYSTEM

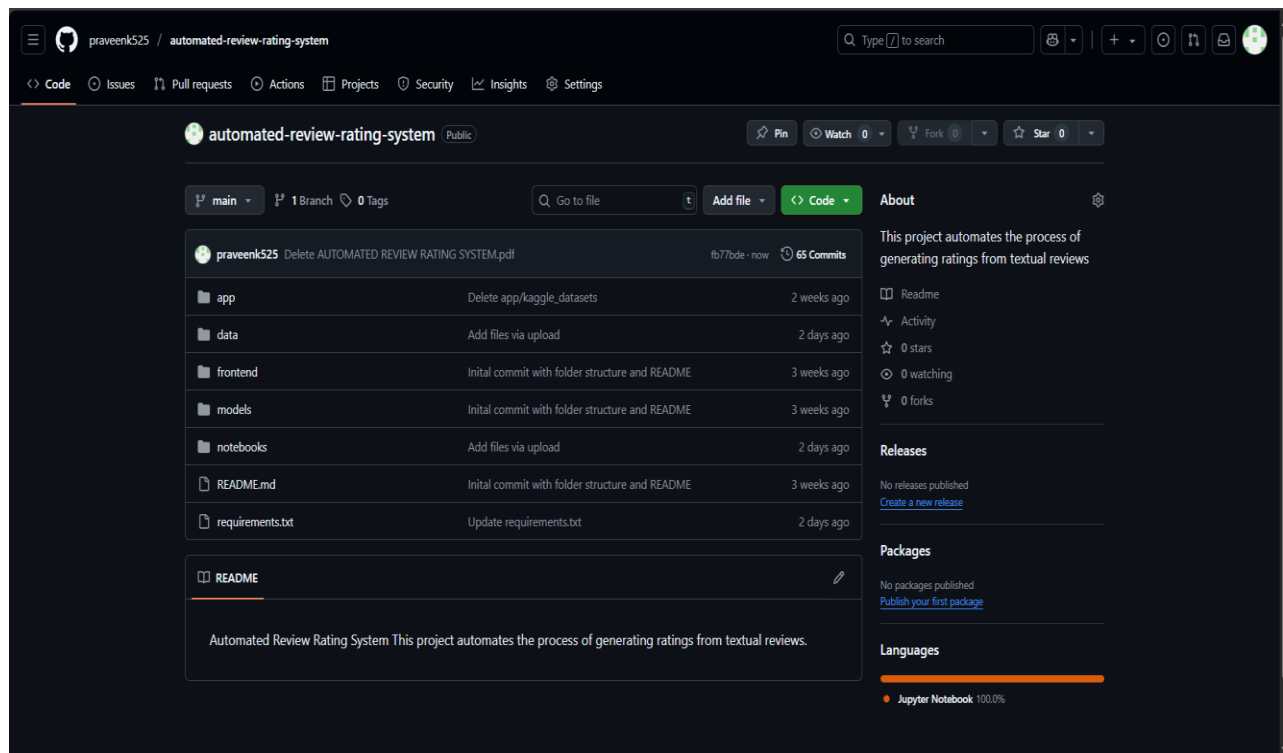
PROJECT OVERVIEW

This project automates the process of generating ratings from textual reviews. an AI-based system that automatically predicts the numerical rating (e.g., 1 to 5 stars) of a product, service, or business based solely on user-generated textual reviews. This aims to reduce rating manipulation, standardize feedback, and assist platforms in quickly gauging sentiment at scale.

ENVIRONMENT SETUP

- Installed python with necessary libraries such as numpy, pandas, matplotlib, scikit-learn, seaborn, nltk
- Colab

GITHUB PROJECT REPOSITORY



DATA COLLECTION

Data was collected from Kaggle datasets:

For the Automated Review Rating System, a total of 10 datasets were collected, including reviews from Amazon in the Electronics and Movies domains. Each dataset contains two main columns: review text and star rating (1–5).

1. Amazon Electronics Reviews

- Contains approximately ~ 6906564 reviews.
- Reviews cover products like phones, laptops, cameras, and other electronics, capturing a wide range of customer sentiments.
- Dataset link: [Amazon-reviews](#)

2. Amazon Movies Reviews

- Contains approximately ~3430888 reviews.
- Includes reviews and ratings of movies and DVDs across various genres, reflecting diverse customer opinions.
- Dataset link: [Amazon Movie Reviews Processed](#)

All datasets are merged into one

DATA PREPROCESSING

Preprocessing is the first and most important step in any machine learning or data science project. It involves cleaning, transforming, and organizing raw data into a format that your machine learning model can understand and learn from effectively.

Load and Inspect Datasets

- Load all 10 datasets (e.g., Women's Clothing, TripAdvisor, etc.)
- Inspect structure: check column names, data types, sample records
- Verify presence of review and rating columns `pd.read_csv()` → Load CSV files into

a DataFrame `pd.read_excel()` → For Excel files

View first few rows

`df1.head()`

```
# View last few rows
```

```
df1.tail()
```

```
# Check column names
```

```
df1.columns
```

```
# Get DataFrame shape (rows, columns)
```

```
df1.shape
```

```
# Check column data
```

```
types df1.dtypes # Full
```

```
summary info df1.info()
```

```
# Summary statistics (for numeric columns)
```

```
df1.describe()
```

Remove Unnecessary Columns

- Drop columns not required for prediction, such as:
 - user_id, product_id, timestamp, title, image_url, etc.
- Keep only relevant columns: usually review_text and rating

Remove Null or Missing Values

- Remove rows with null values in review or rating

```
#Check for missing values in each column
```

```
print(df.isnull().sum())
```

```
# removing null values
```

```
df.dropna(inplace=True)
```

```
# Remove rows where review_text or rating is missing
```

```
df.dropna(subset=['review_text', 'rating'], inplace=True)
```

Remove Duplicates and Conflicting Reviews

- **Exact duplicates:** same review and rating keep one
- **Conflicting reviews:** same review text, different ratings remove all

```
#check duplicates print(df.duplicated().sum()) # Remove all duplicate rows
```

```
df.drop_duplicates(inplace=True)
```

Clean Text Data

- Convert text to lowercase
- Remove:
 - URLs
 - Punctuation
 - Numbers and special characters
 - Extra spaces

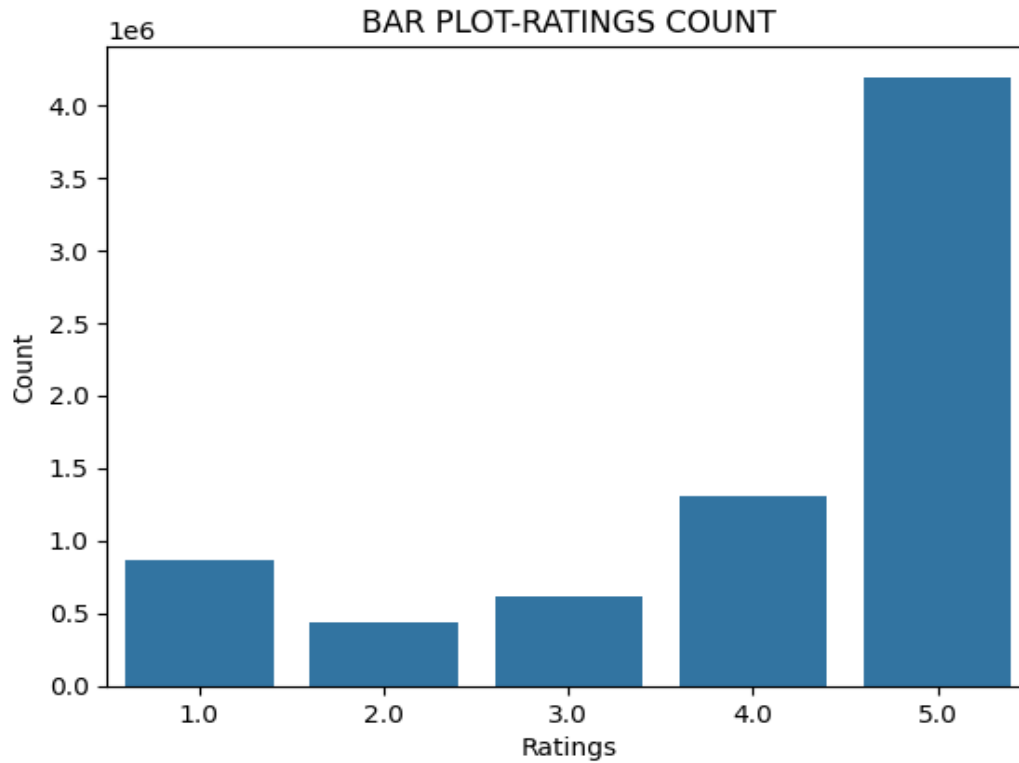
DATA VISUALIZATION

Data Visualization is the process of representing data and information visually using charts, graphs, and other visual elements (like maps or word clouds) to help people understand patterns, trends, and insights in data.

BAR CHART

A bar chart represents categorical data with rectangular bars. The length of each bar is proportional to the value it represents.

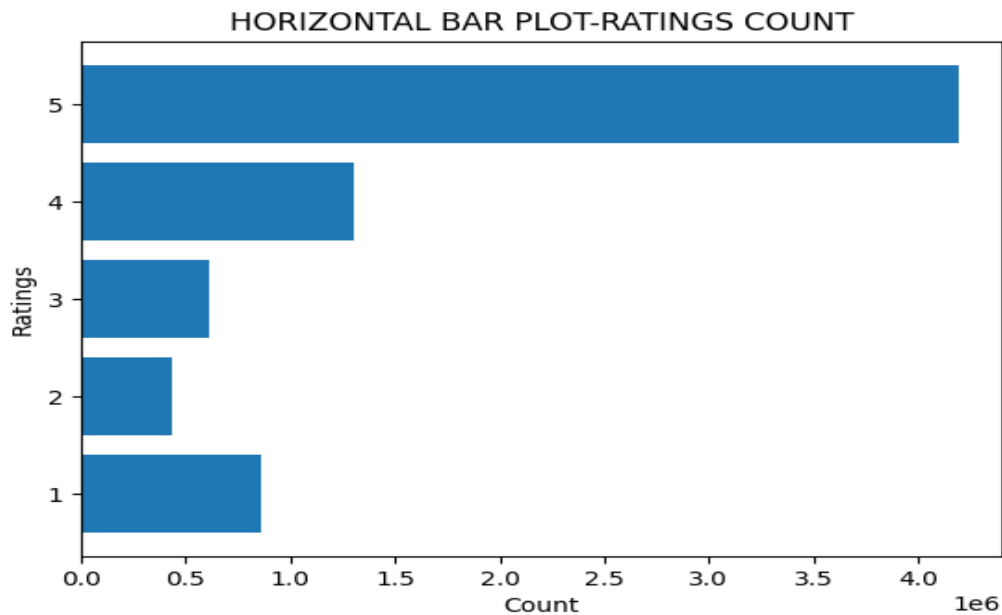
- Use: To show the number of reviews in each star rating (e.g., how many 1-star, 2-star, etc.).
- Number of reviews per rating (1 to 5).



Horizontal Bar Chart (HBar)

A horizontal bar chart is like a bar chart, but the bars extend horizontally instead of vertically.

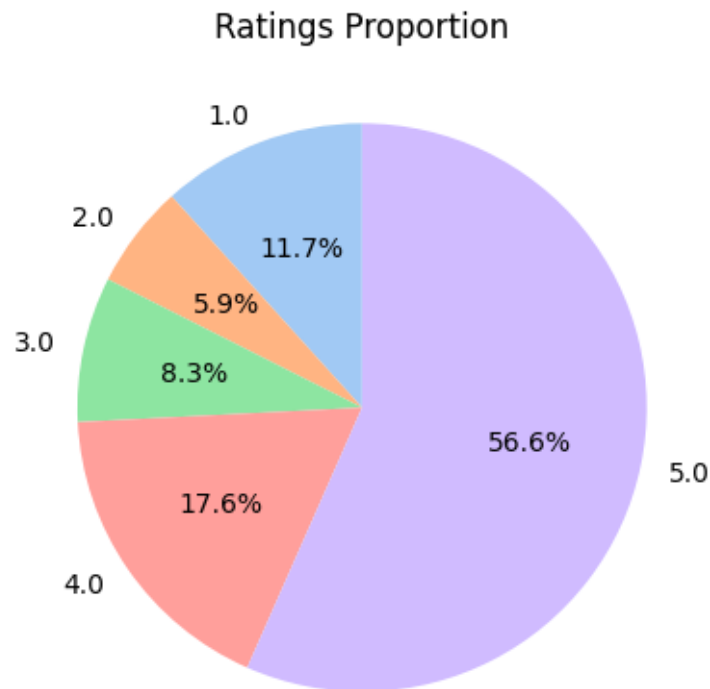
- To show average values like review length for each rating in a clean, readable format.



Pie Chart

A pie chart is a circular chart divided into slices to illustrate proportions of categories.

- To show the percentage of each rating (how much of the data is 5-star, 4-star, etc.).

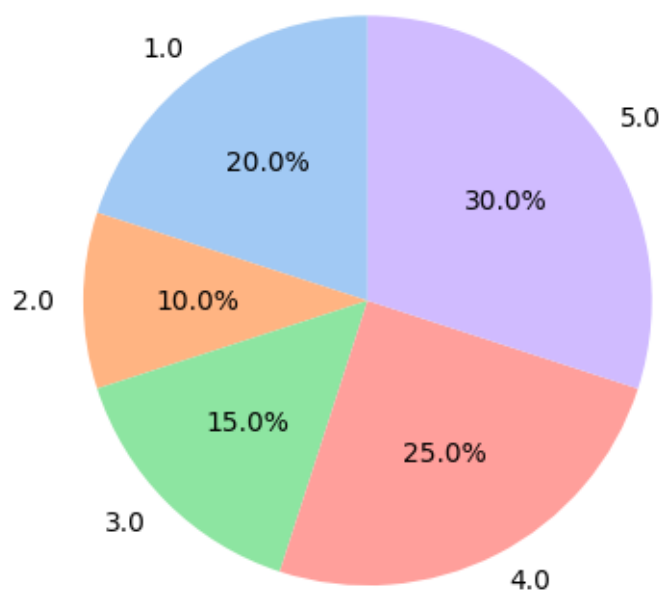


IMBALANCED DATASET

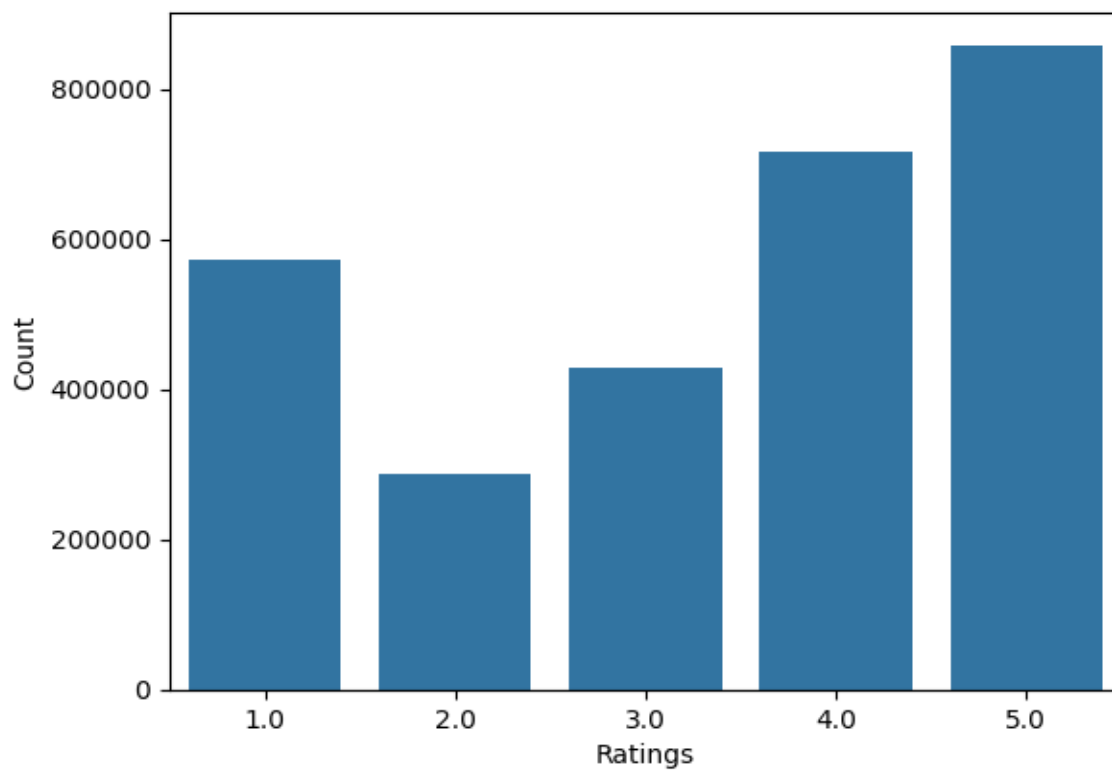
Imbalanced data refers to a dataset in which the distribution of classes is not equal — meaning some classes (labels) have significantly more samples than others.

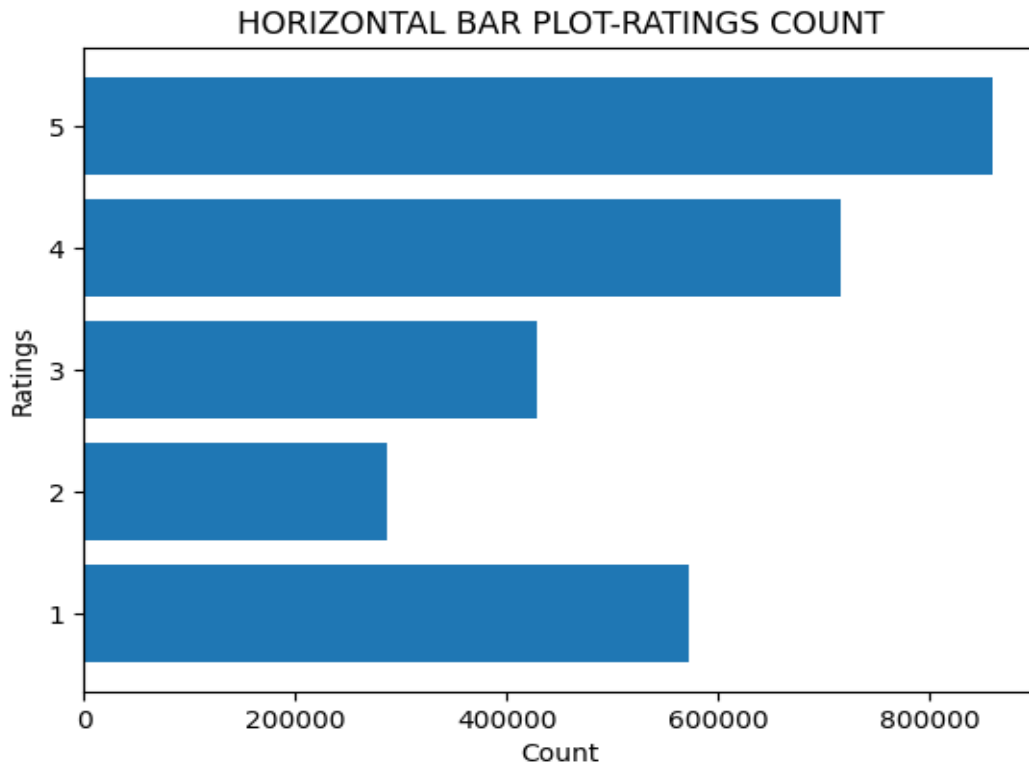
Balanced dataset with ratings: 1-20%, 2-10%, 3-15%, 4-25%, 5-30%

Ratings Proportion



BAR PLOT-RATINGS COUNT





SAMPLES:

Showing 5 sample review(s) for Rating 1:

1. a boring let down when jp kevin bernhardt a playboy club owner buys a mysteriouslooking puzzle box and mistakes it for a work of art he has no idea that pinhead the black pope of hell doug bradley is trapped inside playing tricks on his captors mind pinhead tries his best to manipulate jp into breaking him out of his prison that is until an astute journalist terry farrell begins to suspect something watching hellraiser iii its obvious that the producers were trying to make a poor mans nightmare on elm street rather than a proper continuation of the first two films pinhead spouts bad oneliners like freddy krueger in the lesser elm street entries and a significant part of the premise even involves the hero having dreams about her fathers death the result is a watereddown unsteady film that contains only a small morsel of what made clive barkers first two hellraiser films so interesting substituting tired slasher film ingredients instead a real letdown from the first two films

2. please dont waste your money on them i have had nothing but issues since we received this as a gift cant connect to wifi after several attempts doesnt even prompt you to do so secondly wont recognize my password and after resetting itstill wont recognize after third attempt at resetting itunable to process my request im through with this junk shouldve bough a real tablet when asked

what i preferred for my daughter waste of time and money j u n k take my word for it and dont waste your money i repeat dont waste your money

3. response to chapmans criticism of jadams review j adams may not believe that your comments are worth a response but i cant resist your correct in your selfdescription of a lack of mental capacity i assume that not liking this particular film means that the viewer has no sense of humor wake up this film was a box office flop which according to you logic would mean that most of america has no sense of humor wrong were just tired of seeing the same stupid plot with same stupid jokes and we are tired of hollywood assuming that the public consists of yr old boys who can only understand bathroom humor stupidity does not always equal funny my friends have funnier conversations and it doesnt cost i dont need to throw away money to watch actors being paid millions just to remake their last comedy and i use that term loosely i can turn on cable tv any time of the day and never fail to find another version of this film and if you havent noticed kutchner plays only character and replays the same unfunny jokes each time the comment to see honeymoon in vegas is on the mark at least that film was smart and funny in what happens its like they wrote the script in minutes after viewing kutchers other films if you think elite is expecting an actual plot and humor then i guess most americans are elitists i just hate leaving the theater thinking thats hours of my life that ill never get back if you want to laugh at the silly or slapstick throw in a copy of duck soup

4. directions for installing were clear even without the recommended video instructions but the shiny clear surface takes away from the matte glare free surface of my new kindle paperwhite most aggravating of all the finish tab has a sticky residue that resists all so again my surface on my new kindle is compromised bubbles remain i am unhappy unsatisfied with this product

5. first let me state that i love the color and the swivel the problem i am having with the ipad mini cover is with the grooves at the bottom of the cover that are used to secure the ipad in place when you are typing or playing games the grooves are not deep enough to support the ipad and my ipad is constantly fallen down i am wondering if the cover i received is defective because my friend has a black cover by fintie and the grooves are much deeper than mine it would be nice to have some feed back i would purchase this item again if the company prove to me that i may just have a defective unit

Showing 5 sample review(s) for Rating 2:

1. only over eighteen should see this movie i have seen this movie on tv and although it was good i would never allow my kid to be in such a movie to expose perverts that kidnap a young boy and do despicable things to the child i wont elaborate the word molest says it all it is too bad the movie was made because this is permanent documentation to the young male actor and the adult actors casted in the role of kidnappers this is not a filmography to be proud of the parents of the young boy must of been paid very well to secure the acting services of this young boy i hope any future parent who cares about their children think twice before allowing any of their children to be casted in a similar movie like this addendum i have read some of the criticisms to this review anyway my remarks still stand anytime a person writes or says something that goes outside the boundaries

of accepted opinions of general tribal thinking then you will get flack and even disparaging names if i am to be the only person to give a twostar rating to this movie then so be it here is my response that is short and sweet i have read your letter thank you for sharing now it is behind me god bless and have a nice day

2. after following the directions i was convinced there is another step where i am supposed to remove a thin covering film but there is no such step i am not pleased with the loss of brightness due to this screen

3. it made my macbook air look sleek but the bottom didnt fit at all it fit the sides but there would always be a gap in the middle other than that it was an okay buy but i wouldnt buy it again there are other bettercheaper options than this case

4. the adapter looks a lot smaller than it ends up being if you compare it to the standard adapter the unit looks like it will be a huge space savor i know the lack of space savings is partially my fault in reality if you are on the go a lot and wind up the cords around the adapter instead of neatly bundling them as the picture shows it is almost the same size the bigger issue is this adapter gets very hot i had it draped across a couch and i was concern it was going to damage the fabric due to how hot it was getting i havent used it much since for extended periods of time to rate it on normal to high usage yet but dont leave it on any surface that cant take a lot of heat

5. ive updated my review because while the cover looks nice it is not durable i have cracks on both bottom corners similar to what others have discovered i dont put it in or take it out of a case much but that action plus opening the screen all the way seem to have caused the issue i really wanted to give a good review but franklynnot lasting months without breaking is a design flaw that cant be ignored i am working with them to get a replacement sent to me but franklyill probably just keep it as a backup and buy something differentbetter oh well my original review is below the people at speck have done a great job taking the beauty of the mac air and carefully wrapping it with a sleek and protective cover it adds some slight weight but what cover wont my only longterm concern is how well itll stand up to useabuse the clips are minimal which is good but over the course of a year or two maybe they will start to wear that said i would gladly repurchase this as needed to protect my prized possession but hopefully that wont be necessary enjoy

Showing 5 sample review(s) for Rating 3:

1. my previous router trendnet tewgr was purchased and started acting up recently wireless signal dropping wired connection unstable i was constantly power cycling the router to get my connection back upbr br i decided to replace that router with a linksys even though this particular model doesnt have external antennas i dont have any complaints so farbr br why would you pay for the new retail one when you can save plenty of money on this refurbished modelbr br update br this router started having problems and has been replaced my connection became unreliable and required power cycling the router on a daily basisbr br purchase date br replace date br br as a result i have lowered my review to stars since it was great while it lasted but didnt last as long as i hoped

2. the great thing about this is that it is so fast however it does not allow downloading of itunes so i basically never can use my ipod anymore also i really like the google docs it comes with to do homework but it is almost impossible to turn in my assignments for my online classbr if you are going to buy this buy it at costco i saw it a week after i bought it here but the one at costco is the same price with easy return better warranty and years t mobile service freebr ignoring that though it is a good computer

3. dont bother with it as a film watch only if you like gackt or hyde i happen to like both gackt and hyde not as a fangirl but i like their music and weirdly pretty boys fashion senses i especially appreciate larcenciels music hyde definitely can perform and sing however this movie was a mess it needed better story structure and editing it really was a strange experience in the mishmosh way of telling a vampirehuman bond with homoerotic undertones type of tale hyde plays a vampire who is pretty sick of his life and has a death wish gackt plays a juvenile delinquent in a futuristic asian town who with the aid of his vampire comrade rises in the ranks the fight scenes have a certain overthetop appeal and watching prettyboy gackt and his cheekbones and makeup and sexynotatallpractical outfits for a gansterguy is a bit surreal but amusing hyde has terrible cheap hair in this one what a shame hes really attractive in that uberslender pretty jrock style when his hair is longer and dark the story basically follows the emotional messses and joys of the bond between these two men as they deal with relationships crime ennui dysfunctional families allies and enemies had a better writingediting team gotten ahold of this there might have been a good flick this is a soso flick but definitely a mustsee for fangirls and a couldbegoodsee for those who appreciate two of jrocks cooler bishies mir

4. ibrahead rest mount states it fits any standardsized headrest however i have an suv and the car mount falls off the headrest with the slightest touch it is barely barely hanging onto the headrest i would give this product a star if we didnt run across this important issue i did give it stars solely because the side grips that hold your tablet in place is nicely padded and holds the tablet in place securely until you attach it to your vehicle i like that it has a swivel adjustment so that your little ones can see the screen i would not recommend this product to someone unless they knew for sure it would fit their headrest

5. mixed results my usb lg blueray burner now works and my external usb hard drives work through the hubbr br some of my usb devices are not recognized by my win laptop and some work just fine through the hub

Showing 5 sample review(s) for Rating 4:

1. im enjoying this item
2. love this case thinking of getting a new one since this one is getting a little old
3. great case does its job inexpensive good value for the price love that you can take it apart and use in two ways cons not enough functional when you want to watch video without holding tablet

4. relatively easy to set up with netgear nighthawk ac dual band wifi gigabit router r can now use my ipad android phone laptop etc at the opposite end of my story house which before had little to no signal we now have signal strength when im several feet away from the range extender i like that the extender is small and leaving the other outlet usable the external antennas are more efficient than internal you can adjust the antennas positions for optimal results the connection seems quite stable i recommend it

5. works well with my gopro hero silver

Showing 5 sample review(s) for Rating 5:

1. never a miss with mickey this is as always a sure hit for any child that is a lover of mickey mouse clubhouse my son and daughter love it you cant go wrong with this selection

2. was looking for a cover found this cute polka dot one and bought it love it and use it all the time

3. wife loved it i purchased this dvd movie of the lost kennedy home movies for my wife who enjoys anything about the kennedys especially jackie kennedy onassis she was very pleased with this gift

4. item i received looks like the one in the picture if you look around amazon you will see various brands of these keyboard but most have a cheap keypad you can see the plastic between the keys or just poor quality i even watched a few youtube videos and you can see how horrible the keypad is on some brandsbr br anyways none of those problems with this product keys feel nice and give a nice click when pressed while i havent tested range i havent experienced any disconnects i mostly use it with my android tablet trackpad kb works finebr br model on the sticker is listed asbr rtmwkbbr br prosbr keypad feels nicebr rechargeablebr trackpadbr can rotate the trackpad fn alt if neededbr backlit keys not mentioned in the descriptionbr laser beam is bright and cats go nuts chasing it br works great with android br pricebr battery life seems longbr br consbr rf transmitter wish it was bluetoothbr charging light is confusing it appears red light on second notch is charging when red light goes to the rd notch its fully charged the red led never goes out but hooking it up to a powerbank i can confirm it stops drawing current when light goes to the rd notchbr no caps lock indicator dunno maybe the led lights are just strange on mine

5. fantastic this is the best potty training video ever my children used it and now my grandchildren watch it all the time they learn everything including raising the lid and then putting it back down its much more fun to learn from other kids the songs are great and you will never forget super duper pooper your child will not be disappointed

Train-Test Split

What is Train-Test Split?

In machine learning, we split the dataset into:

- **Training Set** → Used to train the model (e.g., 80% of data)
- **Testing Set** → Used to evaluate the model's performance (e.g., 20% of data)

This helps ensure the model can generalize to unseen data.

Why Use stratify?

If the dataset is imbalanced (e.g., more 5-star reviews than 1-star), a random split could cause:

- Training set to have fewer examples of rare classes.
- Testing set to not represent all classes equally.

Stratification ensures:

- The class distribution (percentage of each rating) is the same in both train and test sets.

Why Bidirectional LSTM was Chosen

The Bidirectional LSTM (BiLSTM) was chosen because:

- **Captures context in both directions:** Unlike standard LSTM, BiLSTM reads the text sequence forward and backward, which is crucial for understanding sentiment in reviews where meaning depends on surrounding words.
- **Effective for sequential data:** Customer reviews are textual sequences; LSTM handles long-term dependencies better than vanilla RNNs.
- **Proven performance:** BiLSTM models achieve higher accuracy in sentiment and review rating prediction tasks compared to traditional ML algorithms or simple neural networks.

Preprocessing Steps Used

Before feeding data into the BiLSTM, the following preprocessing steps were applied:

1. **Text Cleaning:** Converted all text to lowercase and removed special characters and punctuation.
2. **Truncating Excessively Long Reviews:** Reviews longer than 300 words were truncated to reduce noise and ensure uniform input size.
3. **Encoding Labels:** Star ratings (1–5) were label-encoded and converted to one-hot vectors for multi-class classification.

4. **Tokenization and Padding:** Text was tokenized using Keras' Tokenizer and sequences were padded to a fixed length (max_len=200) for uniform input to the model.
5. **Optional Stopword Removal & Lemmatization:** Stopwords were removed and words were lemmatized to reduce vocabulary size and improve semantic understanding.

Model Training Logic

The BiLSTM model was trained using the following logic:

- **Embedding Layer:** Either random or pretrained embeddings (GloVe) were used to convert words into dense vectors.
- **Bidirectional LSTM Layer:** 128 units with dropout and recurrent dropout to capture forward and backward context.
- **Dropout Layer:** 50% dropout applied after BiLSTM to prevent overfitting.
- **Dense Output Layer:** 5 neurons with softmax activation for predicting one of the five star ratings.
- **Loss Function:** categorical_crossentropy for multi-class classification.
- **Optimizer:** Adam optimizer for efficient convergence.
- **Callbacks:** EarlyStopping and ReduceLROnPlateau were used to avoid overfitting and adjust learning rate dynamically.

Training was performed using a stratified 80-20 train-test split to ensure proportional representation of all rating classes.

Evaluation Results and Interpretation

After training, the model was evaluated on the test set:

- **Accuracy:** Measures the proportion of correctly predicted ratings.
- **Precision, Recall, F1-Score:** Evaluated per class to see how well the model predicts each rating.
- **Confusion Matrix:** Visualized misclassifications between ratings.

Accuracy: 0.6417

Classification Report:

	precision	recall	f1-score	support
1	0.70	0.81	0.75	114472
2	0.43	0.22	0.29	57236
3	0.48	0.48	0.48	85854
4	0.60	0.57	0.59	143090
5	0.74	0.81	0.77	171708
accuracy			0.64	572360
macro avg	0.59	0.58	0.58	572360
weighted avg	0.63	0.64	0.63	572360

